

Chapter 3: mock exam problems

Statistics for Insurance in the style of the 2024/25 final exam

0. The single big event computation

In the notes: “if $M_2 = Z_1 \vee Z_2$ then $\mathbb{P}(M_2 > z) = 1 - F(z)^2 = 1 - (1 - \bar{F}(z))^2 \approx 2\bar{F}(z)$ ”. (The middle expression should read $1 - (1 - \bar{F}(z))^2$, with a bar.)

1. **The maximum is below z iff both are below z :** $\{M_2 \leq z\} = \{Z_1 \leq z\} \cap \{Z_2 \leq z\}$.
2. **Independence:** $\mathbb{P}(M_2 \leq z) = \mathbb{P}(Z_1 \leq z)\mathbb{P}(Z_2 \leq z) = F(z)^2$. Hence $\mathbb{P}(M_2 > z) = 1 - F(z)^2$.
3. **Write $F = 1 - \bar{F}$ and expand the square:**

$$1 - (1 - \bar{F}(z))^2 = 1 - (1 - 2\bar{F}(z) + \bar{F}(z)^2) = 2\bar{F}(z) - \bar{F}(z)^2 = 2\bar{F}(z)\left(1 - \frac{\bar{F}(z)}{2}\right).$$

4. **Far in the tail** $\bar{F}(z) \rightarrow 0$, so the bracket tends to 1: $\frac{\mathbb{P}(M_2 > z)}{2\bar{F}(z)} \rightarrow 1$, written $\mathbb{P}(M_2 > z) \approx 2\bar{F}(z)$.

Intuition: the chance that at least one of two claims is huge is about twice the chance for one claim; the chance that *both* are huge ($\bar{F}^2$) is negligible. With n copies, $\mathbb{P}(M_n > z) = 1 - (1 - \bar{F}(z))^n \approx n\bar{F}(z)$ by the binomial expansion. **Subexponential** means the *sum* has the same tail as the *maximum*: $\mathbb{P}(Z_1 + Z_2 > z) \approx \mathbb{P}(M_2 > z) \approx 2\bar{F}(z)$.

1. What the exam looks like, and what that means for Chapter 3

Pattern of the 2024/25 exam

- 3 problems, about 4 subquestions each, all with equal weight. Subquestions can be solved independently using earlier results.
- The subquestion types repeat:
 - (a) a concrete **calculation** (a density, $\mathbb{P}(S = 0)$);
 - (b) **show it belongs to a class** (“show S is in the exponential dispersion family”);
 - (c) **explain in words** (“argue why the simple average is inappropriate”);
 - (d) **prove a limit result** (consistency, a CLT, uniform convergence), usually with a *hint* or a given inequality.
- You must **cite results** (“by Theorem 3.6 ...”).

For Chapter 3 the natural counterparts are: (a) Hill estimates, quantiles, moments, maxima limits; (b) show a distribution is RV, heavy-tailed, or in the Fréchet domain; (c) why k must be intermediate, what the Hill plot shows, why lognormal is not RV; (d) consistency of $\tilde{T}_n$, the Hill variance γ^2 , a bias condition on k , Breiman via Potter + DCT. The problems below cover all of these. Each is written as a full exam problem (4 subquestions) and followed by a solution with references.

2. Problems

Problem 1 (Lomax claims and the Hill estimator)

Let $Z_1, Z_2, \dots$ be iid with $\bar{F}(x) = (1+x)^{-\alpha}$, $x \geq 0$, for an unknown $\alpha > 0$. Let $k = k_n \rightarrow \infty$ with $k/n \rightarrow 0$, $u_n = F^{\leftarrow}(1 - k/n)$, and let $\gamma_{n,k}$ be the Hill estimator (Def. 3.2).

1. Show that $\bar{F}$ is regularly varying, and find γ . For which $p > 0$ is $\mathbb{E}Z_1^p < \infty$?
2. Show that $\gamma = \int_1^\infty T(t) dt/t$ and that $\gamma_{n,k} = \int_1^\infty \hat{T}_n(t) dt/t$.
3. Find u_n explicitly, and show that

$$\int_1^\infty \frac{T_n(t) - T(t)}{t} dt = \frac{1}{\alpha + 1} \left(\frac{k}{n}\right)^\gamma (1 + o(1)).$$

Hint: $\frac{1+ut}{1+u} = t(1+h)$ with $h = -\frac{1-1/t}{u+1}$, and $(1+h)^{-\alpha} = 1 - \alpha h + O(h^2)$. You may interchange limit and integral.

4. Deduce for which sequences k_n condition (3.7) holds. Give the conclusion of Theorem 3.14 and explain the bias-variance trade-off in choosing k .

Problem 2 (the single big event)

Let $Z, Z_1, Z_2, \dots \geq 0$ be iid with distribution function F and $\bar{F}(z) > 0$ for all z .

1. Show that $\mathbb{P}(\max_{i \leq n} Z_i > z) \sim n\bar{F}(z)$ as $z \rightarrow \infty$, for each fixed n .
2. Assume $\bar{F}$ is regularly varying with index $\alpha > 0$. Show that $\mathbb{P}(Z_1 + Z_2 > z) \sim 2\bar{F}(z)$.
3. Let $Z \sim \text{Exp}(1)$. Compute $\mathbb{P}(Z_1 + Z_2 > z)/(2\bar{F}(z))$ and its limit. Explain in words what the difference from 2. means for a reinsurer.
4. In the setting of 2., compute $\lim_{z \rightarrow \infty}$ of: (i) $\mathbb{P}(Z_1 > z \mid Z_1 + Z_2 > z)$, (ii) $\mathbb{P}(Z_1 \vee Z_2 > z \mid Z_1 + Z_2 > z)$, (iii) $\mathbb{P}(Z_1 > z/2, Z_2 > z/2 \mid Z_1 + Z_2 > z)$. Interpret.

Problem 3 (maxima and domains of attraction)

Let $M_n = \max(Z_1, \dots, Z_n)$ for iid $Z_i \sim F$.

1. Let $\bar{F}(x) = x^{-\alpha}$, $x \geq 1$. Show that $\mathbb{P}(M_n/n^{1/\alpha} \leq x) \rightarrow \exp(-x^{-\alpha})$ for every $x > 0$.
2. Let F be continuous with $\bar{F}$ regularly varying with index α , and $a_n = F^{\leftarrow}(1 - 1/n)$. Show that $n\bar{F}(a_n x) \rightarrow x^{-\alpha}$, and deduce $\mathbb{P}(M_n/a_n \leq x) \rightarrow \exp(-x^{-\alpha})$.
3. Let $Z \sim \text{Exp}(1)$. Show that $\mathbb{P}(M_n - \log n \leq x) \rightarrow \exp(-e^{-x})$.
4. Let $\bar{F}(x) = \exp(-\sqrt{x})$. Show that F is heavy-tailed but, using Theorem 3.2, that F is *not* in the Fréchet max-domain of attraction.

Problem 4 (Potter's bounds and Breiman's lemma)

Let $Z \geq 0$ be regularly varying with index α , and $W \geq 0$ independent of Z with $\mathbb{P}(W > 0) > 0$. (Think: Z is the claim size, W a random inflation or exchange-rate factor.)

1. State Potter's bounds, second version (Thm 3.6), and show by an example why $x \geq 1$ cannot be dropped.
2. Assume $W \leq b$ a.s. for a constant b . Show that $\frac{\mathbb{P}(ZW > x)}{\mathbb{P}(Z > x)} \rightarrow \mathbb{E}[W^\alpha]$.
3. Now drop boundedness and assume $\mathbb{E}[W^{\alpha+\varepsilon}] < \infty$ for some $\varepsilon > 0$. Show the same limit.
4. Deduce that ZW is regularly varying with index α . If $\alpha = 2$ and $W \sim U(0, 2)$, how much larger is $\mathbb{P}(ZW > x)$ than $\mathbb{P}(Z > x)$ for large x ?

Problem 5 (tail empirical function and the Hill variance)

Let $\bar{F}$ be regularly varying with index $\alpha = 1/\gamma$, with k, u_n as in Problem 1, so $n\bar{F}(u_n) = k$. Let $\tilde{T}_n(s) = \frac{1}{k} \sum_{i=1}^n \mathbf{1}(Z_i > u_n s)$.

1. Compute $\mathbb{E}\tilde{T}_n(s)$ and $\text{Var}\tilde{T}_n(s)$. Show $\tilde{T}_n(s) \xrightarrow{P} s^{-\alpha}$ for each $s > 0$.
2. Show that $k \text{Cov}(\tilde{T}_n(s), \tilde{T}_n(t)) \rightarrow (s \vee t)^{-\alpha}$.
3. The Hill limit is $X = \int_1^\infty \frac{W(T(t))}{t} dt - \gamma W(1)$ with W a standard Brownian motion. Show that $X = \gamma \left(\int_0^1 \frac{W(v)}{v} dv - W(1) \right)$ and that $\text{Var}(X) = \gamma^2$.
4. Why must $k \rightarrow \infty$ and $k/n \rightarrow 0$? Explain each condition in two sentences, referring to 1. and to Example 3.3.

3. Solutions

Solution 1

1. $\frac{\bar{F}(tx)}{\bar{F}(x)} = \left(\frac{1+x}{1+tx}\right)^\alpha = \left(\frac{1/x+1}{1/x+t}\right)^\alpha \rightarrow t^{-\alpha}$. So the tail is RV with index α and $\gamma = 1/\alpha$ (cf. the shifted Pareto, Example 3.1.2). By Potter's bounds (Thm 3.5) combined with $\mathbb{E}Z^p = \int_0^\infty px^{p-1}\bar{F}(x)dx$: $\mathbb{E}Z^p < \infty$ iff $p < \alpha$. (For $p = \alpha$ directly: $\int_0^\infty x^{\alpha-1}(1+x)^{-\alpha}dx = \infty$.)
2. $\int_1^\infty t^{-1/\gamma-1}dt = [-\gamma t^{-1/\gamma}]_1^\infty = \gamma$. For Hill: $\int_1^\infty \mathbf{1}(Z_i > Z_{n,n-k}t) \frac{dt}{t} = \log^+(Z_i/Z_{n,n-k})$. Summing over i , only the top k order statistics contribute (continuous F , no ties), giving $\frac{1}{k} \sum_{i=1}^k \log(Z_{n,n-i+1}/Z_{n,n-k}) = \gamma_{n,k}$ (Def. 3.2, Section 3.3.2).
3. $\bar{F}(u_n) = k/n \iff (1+u_n)^{-\alpha} = k/n \iff u_n = (n/k)^\gamma - 1$. So $u_n \rightarrow \infty$ and $\frac{1}{u_n+1} = (k/n)^\gamma$. With $u = u_n$: $T_n(t) = \frac{\bar{F}(ut)}{\bar{F}(u)} = \left(\frac{1+ut}{1+u}\right)^{-\alpha} = t^{-\alpha}(1+h)^{-\alpha}$, where $h = -\frac{1-1/t}{u+1}$. Then

$$T_n(t) - T(t) = t^{-\alpha}(-\alpha h + O(h^2)) = \frac{\alpha t^{-\alpha}(1-1/t)}{u+1} + O((u+1)^{-2})t^{-\alpha}.$$

Integrate against dt/t :

$$\frac{\alpha}{u+1} \int_1^\infty (t^{-\alpha-1} - t^{-\alpha-2})dt = \frac{\alpha}{u+1} \left(\frac{1}{\alpha} - \frac{1}{\alpha+1} \right) = \frac{1}{(\alpha+1)(u+1)} = \frac{1}{\alpha+1} \left(\frac{k}{n} \right)^\gamma.$$

The remainder is $O((k/n)^{2\gamma})$, which is o of the main term. (Numerical check with $\alpha = 2$: $u = 100$ gives 0.003325 against $1/303 = 0.003300$.)

4. Condition (3.7) requires $\sqrt{k} \int_1^\infty \frac{T_n - T}{t} dt \rightarrow 0$, i.e.

$$\sqrt{k} (k/n)^\gamma \rightarrow 0 \iff k^{\frac{1}{2}+\gamma} = o(n^\gamma) \iff k = o(n^{2\gamma/(1+2\gamma)}).$$

For example, $\alpha = 2$ gives $k = o(n^{1/2})$. (The condition $\sqrt{k}B_n(s_0) \rightarrow 0$ is handled the same way: $\sup_{t \geq s_0} |T_n - T| = O((k/n)^\gamma)$.) Then Theorem 3.14 gives $\sqrt{k}(\gamma_{n,k} - \gamma) \xrightarrow{d} N(0, \gamma^2)$. *Trade-off*: the variance is $\approx \gamma^2/k$ and *decreases* in k ; the bias is $\approx (k/n)^\gamma/(\alpha+1)$ and *increases* in k , because a lower threshold includes non-Pareto observations. The optimal k balances the two. In practice, choose the largest k in the flat region of the Hill plot.

Solution 2

1. By independence, $\mathbb{P}(M_n > z) = 1 - (1 - \bar{F}(z))^n$. Expand the binomial: $= n\bar{F}(z) - \binom{n}{2}\bar{F}(z)^2 + \dots$. Dividing by $n\bar{F}(z)$ gives $1 + O(\bar{F}(z)) \rightarrow 1$.
2. This is Exercise 1, part A (RV $\Rightarrow$ S). *Lower bound*: $Z_1 + Z_2 \geq M_2$, so the ratio is $\geq \mathbb{P}(M_2 > z)/(2\bar{F}(z)) \rightarrow 1$ by 1. *Upper bound*: for $\delta \in (0, \frac{1}{2})$, $\{Z_1 + Z_2 > z\} \subseteq \{Z_1 > (1-\delta)z\} \cup \{Z_2 > (1-\delta)z\} \cup \{Z_1 > \delta z, Z_2 > \delta z\}$, so

$$\frac{\mathbb{P}(Z_1 + Z_2 > z)}{2\bar{F}(z)} \leq \frac{\bar{F}((1-\delta)z)}{\bar{F}(z)} + \frac{\bar{F}(\delta z)}{\bar{F}(z)} \cdot \frac{\bar{F}(\delta z)}{2} \rightarrow (1-\delta)^{-\alpha} + \delta^{-\alpha} \cdot 0.$$

Let $\delta \downarrow 0$.

3. $Z_1 + Z_2 \sim \Gamma(2, 1)$, so $\mathbb{P}(Z_1 + Z_2 > z) = (1+z)e^{-z}$ and the ratio is $\frac{1+z}{2} \rightarrow \infty$. *In words*: with light tails, a large total usually comes from *many moderate* claims, so diversification works. With heavy tails, a large total comes from *one* huge claim. The reinsurer's risk is then driven by single catastrophic events, and adding more policies does not dilute it.
4. Since $Z_i \geq 0$: $\{Z_1 > z\} \subseteq \{Z_1 + Z_2 > z\}$ and $\{M_2 > z\} \subseteq \{Z_1 + Z_2 > z\}$.

$$(i) \frac{\bar{F}(z)}{\mathbb{P}(Z_1 + Z_2 > z)} \rightarrow \frac{1}{2}.$$

$$(ii) \frac{\mathbb{P}(M_2 > z)}{\mathbb{P}(Z_1 + Z_2 > z)} \rightarrow \frac{2}{2} = 1.$$

$$(iii) \leq \frac{\bar{F}(z/2)^2}{\bar{F}(z)} = \frac{\bar{F}(z/2)}{\bar{F}(z)} \bar{F}(z/2) \rightarrow 2^\alpha \cdot 0 = 0.$$

Interpretation: given a large total, it is (asymptotically certainly) caused by one claim exceeding z on its own, equally likely to be either one, and never by two medium claims.

Solution 3

1. $\mathbb{P}(M_n \leq n^\gamma x) = F(n^\gamma x)^n$. For n large, $n^\gamma x \geq 1$, so $F(n^\gamma x) = 1 - n^{-1}x^{-\alpha}$. Then $(1 - \frac{x^{-\alpha}}{n})^n \rightarrow \exp(-x^{-\alpha})$, the Fréchet distribution (GEV with $\gamma > 0$, Thm 3.1).

2. $a_n \rightarrow \infty$, and by continuity $\bar{F}(a_n) = 1/n$. Then $n\bar{F}(a_n x) = \frac{\bar{F}(a_n x)}{\bar{F}(a_n)} \rightarrow x^{-\alpha}$ by regular variation. Hence

$$\mathbb{P}(M_n \leq a_n x) = \left(1 - \frac{n\bar{F}(a_n x)}{n}\right)^n \rightarrow e^{-x^{-\alpha}},$$

using $(1 - y_n/n)^n \rightarrow e^{-y}$ whenever $y_n \rightarrow y$ (take logs: $n \log(1 - y_n/n) = -y_n + O(y_n^2/n)$). This is the “if” direction of Thm 3.2.

3. $F(x + \log n)^n = (1 - e^{-x - \log n})^n = (1 - \frac{e^{-x}}{n})^n \rightarrow \exp(-e^{-x})$, the Gumbel distribution ($\gamma = 0$).

4. Heavy: $e^{tx}\bar{F}(x) = e^{tx-\sqrt{x}} \rightarrow \infty$ for every $t > 0$, so (3.1) holds (Exercise 4 with $a = \frac{1}{2} < 1$). Not RV: $\frac{\bar{F}(2x)}{\bar{F}(x)} = e^{-(\sqrt{2}-1)\sqrt{x}} \rightarrow 0 \neq 2^{-\alpha}$. By Thm 3.2, the Fréchet domain equals the RV class, so F is not in it (in fact it lies in the Gumbel domain). *Lesson:* heavy-tailed $\nrightarrow$ Fréchet.

Solution 4

1. For each $\varepsilon > 0$ there is C with $\frac{\bar{F}(x/y)}{\bar{F}(x)} \leq C \max(y^{\alpha-\varepsilon}, y^{\alpha+\varepsilon})$ for all $x \geq 1, y > 0$.

Counterexample without $x \geq 1$: take $x = y \rightarrow 0$. Then $\frac{\bar{F}(1)}{\bar{F}(y)} \rightarrow \bar{F}(1) > 0$, whereas $Cy^{\alpha-\varepsilon} \rightarrow 0$.

2. Condition on W (independence; $\{W = 0\}$ contributes nothing):

$$\frac{\mathbb{P}(ZW > x)}{\bar{F}(x)} = \mathbb{E}\left[\frac{\bar{F}(x/W)}{\bar{F}(x)} \mathbf{1}(W > 0)\right].$$

Pointwise: for fixed $W = w > 0$, $\frac{\bar{F}(x/w)}{\bar{F}(x)} \rightarrow w^\alpha$ (RV with $t = 1/w$). *Domination:*

we need a bound that does not depend on x . Potter v2 gives, for all $x \geq 1$, $\frac{\bar{F}(x/w)}{\bar{F}(x)} \leq C \max(w^{\alpha-\varepsilon}, w^{\alpha+\varepsilon}) \leq C \max(1, w)^{\alpha+\varepsilon}$, a constant. By dominated convergence, the limit is $\mathbb{E}[W^\alpha]$.

3. Same argument. Now the dominating variable $C \max(W^{\alpha-\varepsilon}, W^{\alpha+\varepsilon}) \leq C(1 + W^{\alpha+\varepsilon})$ is integrable by assumption, so DCT still applies. This is Breiman’s lemma (Exercise 7 of the notes).

4. Write $H(x) = \mathbb{P}(ZW > x)$ and $c_W = \mathbb{E}W^\alpha \in (0, \infty)$ (positive since $\mathbb{P}(W > 0) > 0$). Then

$$\frac{H(tx)}{H(x)} = \underbrace{\frac{H(tx)}{\bar{F}(tx)}}_{\rightarrow c_W} \cdot \underbrace{\frac{\bar{F}(tx)}{\bar{F}(x)}}_{\rightarrow t^{-\alpha}} \cdot \underbrace{\frac{\bar{F}(x)}{H(x)}}_{\rightarrow 1/c_W} \rightarrow t^{-\alpha}.$$

So multiplying by an independent, lighter-tailed factor keeps the tail index. For $W \sim U(0, 2)$, $\alpha = 2$: $\mathbb{E}W^2 = \int_0^2 \frac{w^2}{2} dw = \frac{4}{3}$, so $\mathbb{P}(ZW > x) \approx \frac{4}{3}\mathbb{P}(Z > x)$, i.e. about 33% more tail probability.

Solution 5

1. $\tilde{T}_n(s)$ is $\frac{1}{k} \times \text{Binomial}(n, p_s)$ with $p_s = \bar{F}(u_n s)$. So $\mathbb{E}\tilde{T}_n(s) = \frac{np_s}{k} = \frac{\bar{F}(u_n s)}{\bar{F}(u_n)} = T_n(s) \rightarrow s^{-\alpha}$ (using $n/k = 1/\bar{F}(u_n)$, $u_n \rightarrow \infty$), and

$$\text{Var } \tilde{T}_n(s) = \frac{np_s(1-p_s)}{k^2} = \frac{1}{k} \cdot \frac{\bar{F}(u_n s)}{\bar{F}(u_n)} (1-p_s) \rightarrow 0 \cdot s^{-\alpha} = 0.$$

By Chebyshev, $\tilde{T}_n(s) - T_n(s) \xrightarrow{P} 0$, and $T_n(s) \rightarrow s^{-\alpha}$. Hence $\tilde{T}_n(s) \xrightarrow{P} s^{-\alpha}$ (Thm 3.9).

2. The indicators for different i are independent, so only same- i covariances count:

$$k \text{ Cov} = \frac{n}{k} [\bar{F}(u_n(s \vee t)) - \bar{F}(u_n s) \bar{F}(u_n t)] = \frac{\bar{F}(u_n(s \vee t))}{\bar{F}(u_n)} - \frac{\bar{F}(u_n s)}{\bar{F}(u_n)} \bar{F}(u_n t) \rightarrow (s \vee t)^{-\alpha} - s^{-\alpha} \cdot 0.$$

(This is the covariance of $W(T(\cdot))$ in Thm 3.10, since $\text{Cov}(W(a), W(b)) = a \wedge b$ and $T(s) \wedge T(t) = T(s \vee t)$.)

3. Substitute $v = T(t) = t^{-1/\gamma}$, so $t = v^{-\gamma}$ and $\frac{dt}{t} = -\gamma \frac{dv}{v}$; $t : 1 \rightarrow \infty$ becomes $v : 1 \rightarrow 0$. Then $\int_1^\infty \frac{W(T(t))}{t} dt = \gamma \int_0^1 \frac{W(v)}{v} dv$, which gives the form stated. Let $I = \int_0^1 W(v)/v dv$. Using $\mathbb{E}[W(a)W(b)] = a \wedge b$:

$$\mathbb{E}I^2 = \int_0^1 \int_0^1 \frac{v \wedge w}{vw} dv dw = \iint \frac{1}{v \vee w} = 2 \int_0^1 \int_0^w \frac{1}{w} dv dw = 2,$$

$$\mathbb{E}[I W(1)] = \int_0^1 \frac{v}{v} dv = 1, \quad \mathbb{E}W(1)^2 = 1.$$

Hence $\text{Var}X = \gamma^2(2 - 2 \cdot 1 + 1) = \gamma^2$. This matches $N(0, \gamma^2)$ in Thm 3.14.

4. $k \rightarrow \infty$: by 1., $\text{Var } \tilde{T}_n(s) \approx s^{-\alpha}/k$. With k fixed the variance does not vanish, since only k observations are used, and neither $\tilde{T}_n$ nor Hill is consistent. $k/n \rightarrow 0$: this makes $u_n \rightarrow \infty$, so that $T_n(s) = \bar{F}(u_n s)/\bar{F}(u_n) \rightarrow s^{-\alpha}$. With a fixed threshold (Example 3.3), $T_n(s)$ stays at $s^{-1/\gamma} \frac{1+\log u}{1+\log(us)} \neq s^{-1/\gamma}$: the estimator converges, but to the wrong thing (bias).

4. Other short questions that could appear

- Show that the lognormal is heavy-tailed but not RV (Example 3.2). The key computations are the mgf integral and Mills' ratio.
- Show that $\mathbb{E}[\log(Z/u) \mid Z > u] \rightarrow \gamma$ (Potter v2 + DCT; see the Potter guide, Use 5).
- Derive Weissman's quantile estimator $\hat{q}_p = Z_{n,n-k} \left(\frac{k}{n(1-p)} \right)^{\hat{\gamma}}$ from $\bar{F}(x) \approx \frac{k}{n} (x/Z_{n,n-k})^{-1/\gamma}$.
- Compute the Hill estimate by hand for a small dataset and give a confidence interval.
- Explain what a Hill plot is, and how you choose k from it.
- Show Z^p is RV with index α/p ; show the mixture or maximum of two RV variables has the smaller index.